

MUHAMMAD ADEEL

Data Scientist

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SUMMARY

Data Analyst transitioning into Data Science, with hands-on experience in statistical analysis, predictive modeling, and machine learning pipelines across healthcare imaging, NLP, and ensemble modeling projects. Proficient in Python, SQL, and end-to-end ML workflows — data preprocessing, feature engineering, model training, and evaluation — with growing exposure to model deployment and applied statistics.

CORE SKILLS

Languages & Databases: Python, SQL, MySQL

Data Analysis & Visualization: Pandas, NumPy, Excel (Pivot Tables, VLOOKUP, SUMIFS), Matplotlib

Machine Learning: Regression, Classification, Ensemble Methods (XGBoost), Scikit-learn, Precision/Recall, Confusion Matrix, Cross-Validation

Deep Learning: TensorFlow, PyTorch, CNN, RNN, GNN, DenseNet121, EfficientNetV2, YOLO, BLIP, Transfer Learning

Computer Vision & Audio: Image Classification, Object Detection, Image Captioning, OCR, Data Augmentation

Deployment & Tools: Flask, Git/GitHub

PROFESSIONAL EXPERIENCE

Data Analyst — Positive Solutions, Wah Cantt

Apr 2026 – Jul 2026

- Analyzed datasets using Excel, MySQL, and Python to support recurring business reporting across departments
- Built and maintained dashboards to track key performance metrics and surface trends for stakeholders
- Wrote and optimized SQL queries (joins, aggregations, subqueries) to extract and clean data from company databases, improving reporting accuracy
- Automated repetitive data-processing tasks using Python, reducing manual effort and supporting faster decision-making

Python Development Trainee — NAVTTC (Training), Islamabad

Oct 2025 – Mar 2026

- Assisted in developing and testing ML models; worked extensively with NumPy and Pandas for data preprocessing, cleaning, and visualization to support model training and evaluation

PROJECTS

Spine Disease Detection System Using EfficientNetV2 — Final Year Project

- Developed a computer vision system to detect and classify spinal disorders (Neural Foraminal Narrowing, Subarticular Stenosis, Spinal Canal Stenosis) using the RSNA 2024 Lumbar Spine dataset
- Fine-tuned EfficientNetV2, applying augmentation and sampling techniques to address class imbalance; deployed via Flask, achieving 95% classification accuracy

AI-Powered “Third Eye” for Visually Impaired Persons

- Built a real-time multi-module assistive system combining DenseNet121 (currency classification), YOLO11n (object detection), and BLIP (scene captioning), with OCR and text-to-speech on a Pakistan-specific dataset
- Achieved 90% classification accuracy, enabling real-time spoken feedback for visually impaired users to interpret surroundings, text, and currency

Heart Disease Prediction — Ensemble Machine Learning

- Built and compared classification models (Logistic Regression, Random Forest, SVM, XGBoost, LightGBM, CatBoost) using Scikit-learn on the Kaggle Playground Series S6E2 dataset
- Achieved an ensemble AUC of ~0.956 on binary classification by averaging model probabilities, with XGBoost delivering the strongest individual performance

CERTIFICATIONS

- Machine Learning Specialization — Andrew Ng (Coursera) — [Verify here](#)
- Deep Learning Specialization — Andrew Ng (Coursera) — [Verify here](#)
- Python Data Structures — University of Michigan (Coursera)

EDUCATION

Bachelor in Computer Science — University of Haripur

Sep 2021 – Jun 2025

- Final Year Project: Spine Disease Detection System ([see Projects](#))

Faculty of Computer Science — FG Science Degree College, Wah Cantt

2019 – 2021

HONOURS & AWARDS

- Secured 3rd Position (Overall, Department of Information Technology), University of Haripur — Final Year Project: Spine Disease Detection System

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Cricket Team Captain, University of Haripur — Led team strategy and player coordination across matches
- Core Member, Tech Society, University of Haripur — Helped organize technical events and workshops

CONFERENCES & SEMINARS

- Machine Learning Bootcamp — University of Haripur | The Robotics with AI Seminar